



Euler's Phi Algorithm

Euler's phi algorithm is a method for computing Euler's totient function, which counts the number of positive integers up to a given integer n that are relatively prime to n .

The algorithm is based on the fact that if n is a product of distinct prime factors $p_1, p_2, \dots, p_k$, then

$$\phi(n) = (p_1-1) * (p_2-1) * \dots * (p_k-1) * p_1^{(k_1-1)} * p_2^{(k_2-1)} * \dots * p_k^{(k_k-1)},$$

where $k_1, k_2, \dots, k_k$ are the powers to which the prime factors appear in the prime factorization of n .

Therefore, the algorithm proceeds as follows:

1. Factorize n into its distinct prime factors, with their corresponding powers.
2. For each prime factor p , compute $p^{(k-1)}$ and p^k , where k is the power of p in the prime factorization of n .
3. Compute the product of all $(p-1)*p^{(k-1)}$ for each prime factor p .
4. Return the product from step 3 as the value of Euler's totient function $\phi(n)$.

Here is an example of applying the algorithm to compute $\phi(24)$:

1. Factorize $24 = 2^3 * 3^1$.
2. Compute 2^2 and 2^3 , and compute 3^0 and 3^1 .
3. Compute $(2)^1 * (3-1)$
4. Return 8 as the value of $\phi(24)$.

Therefore, $\phi(24) = 8$.

let's take another example to illustrate Euler's phi algorithm.

Let's say we want to compute the value of Euler's totient function $\phi(60)$.

1. Factorize $60 = 2^2 * 3^1 * 5^1$.
2. Compute $2^1, 2^2, 3^0, 3^1, 5^0$, and 5^1 .
3. Compute $(2-1)*2^1 * (2-1)*2^2 * (3-1)*3^0 * (5-1)*5^0 * (5-1)*5^1 = 1*2*2*4*4 = 32$.
4. Return 32 as the value of $\phi(60)$.

Therefore, $\phi(60) = 32$.

In this example, we first factorized 60 into its distinct prime factors: $2^2, 3^1$, and 5^1 . Then, we computed the necessary powers of each prime factor. Finally, we used the formula $(p-1)*p^{(k-1)}$ to calculate the contribution of each prime factor to the value of Euler's totient function. We multiplied all these contributions together to get the final value of $\phi(60)$.

```
1. import java.util.*;
2.
3. public class EulerPhiAlgorithm {
4.
5.     // Returns the value of Euler's totient function phi(n)
6.     public static int phi(int n) {
7.         int result = n; // Initialize result as n
8.
9.         // Check for all prime factors of n and subtract their multiples from
        result
10.        for (int p = 2; p * p <= n; p++) {
11.            if (n % p == 0) { // p is a prime factor of n
12.                while (n % p == 0) { // Remove all multiples of p from n
13.                    n /= p;
14.                }
15.                result -= result / p;
16.            }
17.        }
```

```
1.         // If n has a prime factor greater than sqrt(n), then add its
           contribution
2.         if (n > 1) {
3.             result -= result / n;
4.         }
5.         return result;
6.     }
7.     // Main method to test the program
8.     public static void main(String[] args) {
9.         Scanner sc = new Scanner(System.in);
10.        System.out.print("Enter the value of n: ");
11.        int n = sc.nextInt();
12.        int phi_n = phi(n);
13.        System.out.println("phi(" + n + ") = " + phi_n);
14.        sc.close();
15.    }
16.}
```



THANK YOU